

Submit Prediction

Predict the cumulative grain size distribution of a soil sample from photographs of its surface.

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Files

165 files

Size

395.38 MB

Type

jpg, csv

License

Attribution 4.0 International (CC ...)

What files do you need:

train.csv — the training labels: cumulative grain size distributions of the 24 training soil samples. Each row represents one soil sample; each sample is also represented by multiple photographs in the image folder, where the file naming links the images to the sample ID in this CSV.

test.csv — the test set: list of sample IDs for which predictions must be submitted. The corresponding grain size distributions are not disclosed.

sample_submission.csv — a sample submission file in the correct format. It contains all required test sample IDs and a placeholder distribution.

All_Photos_training/test —two folder containing the photographs of each soil sample. The image file names link each photograph to its sample_id. The camera with which the photo was taken is indicated in the name of each photo

ppm.csv — contains the pixel-per-millimeter (PPM) resolution and technical specifications of each camera used in the dataset. The camera

Predicting Soil Grain Size Distributions from Images

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ppm.csv — contains the pixel-per-millimeter (PPM) resolution and technical specifications of each camera used in the dataset. The camera used for each photo is indicated in the photo name itself

The grain size distribution is described as a cumulative curve at 11 fixed grain diameters defined by DIN EN ISO 14688-1. For each diameter, the value indicates the percentage of the soil mass that consists of particles smaller than that diameter. By definition, the curve is monotonically non-decreasing, bounded between 0 % and 100 %, and equals 100 % at the largest diameter (200 mm).

Columns

sample_id — Unique identifier of the soil sample (e.g. F827, G190, H030). The same ID is used in the file names of the photographs that belong to this sample.

Properties of the distribution Each row is a cumulative distribution, so the values are non-decreasing across the 11 grain-size columns and bounded between 0 % and 100 %. By definition, the value at the largest grain size (200 mm) equals 100 %.

Origin of the labels

The raw grain size distributions originate from sieve and sedimentation analyses of the corresponding soil samples. Because these laboratory measurements are not taken at standardized diameters, the raw curves were interpolated linearly on a logarithmic grain-size axis ($\log_{10}(d)$) onto the 11 fixed support points.

Submission target

See sample_submission.csv for the required format.

Submission file

For each `sample_id` in the test set, predict the eleven cumulative mass fractions of the grain size distribution. The submission file must



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Submission file

For each `sample_id` in the test set, predict the eleven cumulative mass fractions of the grain size distribution. The submission file must contain a header and follow this format:

```
sample_id,0.002,0.0063,0.02,0.063,0.2,0.63,2,6.3,20,63,200
TEST_01,1.5,4.2,9.8,21.3,42.7,61.4,75.9,86.2,93.5,98.1,100.0
TEST_02,0.8,3.1,9.7,22.3,44.6,63.1,78.9,88.4,95.1,99.0,100.0
...
```

Each row represents the predicted grain size distribution of one test sample. Predictions must satisfy the following rules:

- one row per `sample_id` in `test.csv`, no extras and none missing
- all eleven grain-size columns present, with the exact header names shown above
- values numeric and in the range [0, 100]
- monotonically non-decreasing across the eleven columns (cumulative distribution)
- exactly 100 at the final column (200 mm)

Submissions that violate any of these rules will be rejected by the scoring system with an error message indicating the issue.'

`## Columns

- `sample_id` - unique identifier of the soil sample (e.g. F827, G190, H030); the same ID appears in the file names of the corresponding photographs



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- `sample_id` - unique identifier of the soil sample (e.g. F827, G190, H030); the same ID appears in the file names of the corresponding photographs
- `0.002` - cumulative mass fraction in percent (0–100) of soil finer than 0.002 mm (clay / fine silt boundary, DIN EN ISO 14688-1)
- `0.0063` - cumulative mass fraction in percent (0–100) of soil finer than 0.0063 mm (fine silt / medium silt boundary)
- `0.02` - cumulative mass fraction in percent (0–100) of soil finer than 0.02 mm (medium silt / coarse silt boundary)
- `0.063` - cumulative mass fraction in percent (0–100) of soil finer than 0.063 mm (coarse silt / fine sand boundary)
- `0.2` - cumulative mass fraction in percent (0–100) of soil finer than 0.2 mm (fine sand / medium sand boundary)
- `0.63` - cumulative mass fraction in percent (0–100) of soil finer than 0.63 mm (medium sand / coarse sand boundary)
- `2` - cumulative mass fraction in percent (0–100) of soil finer than 2 mm (coarse sand / fine gravel boundary)
- `6.3` - cumulative mass fraction in percent (0–100) of soil finer than 6.3 mm (fine gravel / medium gravel boundary)
- `20` - cumulative mass fraction in percent (0–100) of soil finer than 20 mm (medium gravel / coarse gravel boundary)
- `63` - cumulative mass fraction in percent (0–100) of soil finer than 63 mm (coarse gravel / cobble boundary)
- `200` - cumulative mass fraction in percent (0–100) of soil finer than 200 mm (always equals 100 by definition)

Training_labels_updated.csv (2.07 kB)



Data Explorer

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Training_labels_updated.csv (2.07 kB)



Detail Compact Column

10 of 12 columns

sample_id	# 0.002	# 0.0063	# 0.02	# 0.063	# 0.2
24 unique values					
F827	9.4904	19.3892	47.6257	89.5554	99.896
G190	5.2076	10.2425	23.7537	50.3113	75.831
H030	6.7833	10.9091	19.0599	37.3235	66.834
H037	0.932	6.45	16.3216	27.8019	34.662
H038	1.8218	5.14	10.6113	18.2778	24.642
H126	6.333	12.5995	32.5463	68.3599	84.635
H181	0.0	0.0	0.0	4.2956	7.8382
H183	0.0	0.0	0.0	4.9489	8.1046
H366	0.9633	2.0893	5.0347	9.4833	13.748
H067	0.0000	1.0504	1.1400	7.5417	11.000

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- Test_All_Photos
- Training-All_Photos_updated
- Training_labels_updated.csv**
- ppm_updated.csv
- sample_submission.csv

Summary

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H037	0.932	6.45	16.3216	27.8019	34.662
H038	1.8218	5.14	10.6113	18.2778	24.642
H126	6.333	12.5995	32.5463	68.3599	84.635
H181	0.0	0.0	0.0	4.2956	7.8382
H183	0.0	0.0	0.0	4.9489	8.1046
H366	0.9633	2.0893	5.0347	9.4833	13.748
H367	0.8232	1.8584	4.1483	7.5417	11.099
H368	0.5878	1.2731	2.7489	4.9809	7.9203
H371	1.1625	2.4517	5.8097	8.9449	12.792
H372	1.2023	2.3664	5.6877	8.6936	12.172
H374	1.1223	2.4374	5.5348	9.3031	13.126
H405	17.7068	26.4625	47.111	75.656	91.341

- ▶ 165 files
- ▶ 29 columns

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Metadata